

Sleep Duration vs Productivity

Sleep Duration vs Productivity

Team Members: Prageeth Jayasooriya : S/20/398
Menuka Madhuranga : S/20/496
Mohamed Naflan : S/20/463
Dunidu Randunuge : S/20/497
Chathuranga Wijesingha : S/20/559

Abstract

This study develops a mathematical model to investigate how sleep duration and demographic factors influence productivity. A Productivity Index (PI) was defined using measurable factors, including physical activity and stress levels. Data were collected from the dataset of Kaggle.com, with sleep duration and sleep quality treated as numerical variables, and gender as a categorical variable. A linear regression model was fitted using a logit-transformed productivity index as the response, incorporating a quadratic term for sleep duration to account for non-linear effects. The analysis identified gender, sleep duration, and sleep quality as significant predictors of productivity. The model exhibited a good fit, with an R-squared of 0.711, an Adjusted R-squared of 0.708, and a residual standard error of 5.79, indicating that these factors explain a substantial portion of variation in productivity. These results highlight the importance of sleep optimization and demographic considerations in enhancing productivity and provide a foundation for further research on lifestyle factors affecting work performance.

Introduction

Productivity plays a central role in personal, educational, and organizational outcomes, influencing efficiency, performance, and overall well-being. Identifying the factors that enhance or reduce productivity is crucial for developing strategies that improve individual performance and support healthy lifestyles. While multiple variables contribute to productivity, lifestyle factors such as sleep patterns, stress levels, and physical activity have been repeatedly highlighted in research as significant determinants. Adequate sleep improves cognitive function, alertness, memory, and decision-making, whereas insufficient or poor-quality sleep can lead to reduced concentration, slower reaction times, and lower work efficiency. Stress, both physical and psychological, negatively affects focus, energy, and motivation, while regular physical activity has been shown to boost energy levels, enhance mental resilience, and indirectly improve productivity.

Despite the acknowledged importance of these variables, there is limited research that systematically combines sleep, stress, physical activity, and demographic factors into a quantitative framework for measuring productivity. Many existing studies examine these factors in isolation or focus on subjective outcomes without defining a standardized productivity metric. This study addresses this gap by developing a Productivity Index (PI) that integrates measurable aspects of physical activity, stress levels, and sleep characteristics, offering a unified measure for evaluating individual productivity. By applying a linear regression model with a logit-transformed productivity index, this study accounts for the bounded nature of productivity scores and allows for reliable statistical inference. Quadratic terms are included for sleep duration to capture potential non-linear effects, as previous research suggests that both too little and excessive sleep may reduce productivity.

The data for this study were collected from 374 participants in Kaggle.com, including both numerical variables such as sleep duration and sleep quality, and categorical variables such as gender and occupation. The model evaluates the impact of these variables on the productivity index and identifies which factors significantly influence productivity outcomes. Understanding these relationships provides practical insights into how individuals can optimize their daily routines and lifestyle choices to improve productivity. Moreover, the findings have implications for organizations and policymakers seeking to design interventions that enhance performance while promoting health and well-being.

Overall, this study contributes to the growing body of research on productivity by offering a quantitative, data-driven approach that links lifestyle and demographic factors to measurable productivity outcomes. The results highlight key predictors and offer guidance on how targeted changes in sleep, activity, and stress management can enhance individual performance and efficiency.

Data Description

The dataset used in this study consists of 374 participants from Kaggle.com, capturing both demographic and lifestyle variables to model productivity. The variables considered include numerical measures of sleep, activity, and stress, as well as categorical demographic information. A brief description of the variables is as follows:

- **Gender:** The biological sex of the individual, categorized as Male or Female. This variable allows the model to assess potential differences in productivity associated with gender.
- **Sleep Duration:** The average number of hours slept per day, reported by each participant. Sleep duration is a numerical variable and is included in the model with a quadratic term to capture potential non-linear effects, as both insufficient and excessive sleep may influence productivity.
- **Quality of Sleep:** A self-reported measure on a scale of 1 to 10, where higher values indicate better sleep quality. This numerical variable accounts for subjective aspects of sleep that may impact productivity.
- **Physical Activity Level:** The average daily physical activity measured in minutes. This numerical variable reflects the participant's level of energy expenditure and has been included to assess its effect on the productivity index.
- **Stress Level:** A self-reported measure on a scale of 1 to 10, with higher values indicating higher stress. This numerical variable captures the psychological factor that can negatively affect productivity.

The Productivity Index (PI) was defined based on Physical activity Level and the Stress Level.

Methodology

This study followed a two-stage modeling framework to develop and analyze a Productivity Index (PI). In the first stage, a logistic regression model was used to define the productivity index based on physical activity and stress level. In the second stage, a multiple linear regression model was fitted using demographic and sleep-related factors to identify significant predictors of productivity.

1. Development of the Productivity Index (PI)

To quantify individual productivity, a **logistic regression model** was first developed using **Physical Activity Level** and **Stress Level** as predictors. Productivity was represented as a **binary outcome variable**, where individuals were classified as *productive* or *not productive* based on following criteria;

$$\text{Productive} = \begin{cases} 1, & \text{if Physical Activity Level} \geq 50 \text{ minutes and Stress Level} \leq 5 \\ 0, & \text{otherwise} \end{cases}$$

The logistic regression model was specified as

$$\text{logit}(p) = \ln \left(\frac{p}{1-p} \right) = \beta_0 + \beta_1(\text{Physical Activity Level}) + \beta_2(\text{Stress Level})$$

The fitted logistic regression model produced the following estimates:

Predictor	Estimate	Std. Error	z-value	p-value	Significance
Intercept	1.15098	0.97328	1.183	0.237	—
Physical Activity Level	0.31911	0.05068	6.296	3.05e-10	***
Stress Level	-4.24669	0.70923	-5.988	2.13e-09	***

The model yielded a Residual Deviance of 147.05 on 371 degrees of freedom and an AIC of 153.05, indicating good model fit. Both *Physical Activity Level* and *Stress Level* were statistically significant ($p < 0.001$), showing that higher physical activity increases productivity, while higher stress levels decrease productivity.

The predicted probabilities (P_i) were transformed into log-odds (logit values) to overcome the bounded range (0–1) of probabilities. These logit-transformed values were used as the Productivity Index (PI):

$$PI_i = \ln \left(\frac{p_i}{1-p_i} \right)$$

This continuous, unbounded index represents the overall productivity tendency of each individual based on physical and psychological factors.

2. Modeling the Productivity Index Using Linear Regression

In the second stage, the logit-transformed Productivity Index (PI) was modeled using a multiple linear regression framework to evaluate the effect of demographic and sleep-related variables. The independent variables included Gender, Sleep Duration, and Quality of Sleep. A quadratic term for Sleep Duration was added to capture the potential non-linear relationship between sleep and productivity.

The final model is given by:

$$PI_i = \beta_0 + \beta_1(\text{Gender}_i) + \beta_2(\text{Sleep Duration}_i) + \beta_3(\text{Sleep Duration}_i^2) + \beta_4(\text{Quality of Sleep}_i) + \varepsilon_i$$

where ε_i denotes the random error term assumed to follow a normal distribution with mean zero and constant variance.

The summary of the fitted model is shown below:

<i>Predictor</i>	<i>Estimate</i>	<i>Std. Error</i>	<i>t-value</i>	<i>p-value</i>	<i>Significance</i>
<i>(Intercept)</i>	-355.2080	30.6752	-11.580	< 2e-16	***
<i>Gender (Male)</i>	-4.5788	0.7008	-6.533	2.14e-10	***
<i>Sleep Duration</i>	88.2942	8.8572	9.969	< 2e-16	***
<i>Sleep Duration²</i>	-5.8185	0.6037	-9.638	< 2e-16	***
<i>Quality of Sleep</i>	3.3449	0.6019	5.557	5.25e-08	***

All predictors in the model were statistically significant at the 1% level, indicating that each factor plays a meaningful role in determining productivity.

The model achieved a Multiple R-squared of 0.7111 and an Adjusted R-squared of 0.708, suggesting that approximately 71% of the variation in the productivity index is explained by the included variables. The F-statistic (227.1, $p < 2.2e-16$) confirms that the model as a whole is highly significant.

The interpretation of the coefficients provides valuable insights:

- The negative coefficient for Gender (Male = -4.5788) suggests that, on average, males have a lower productivity index compared to females when other variables are held constant.

- The positive linear term (88.2942) and negative quadratic term (-5.8185) for Sleep Duration indicate a curved relationship, where productivity increases with sleep up to an optimal level and then begins to decline with excessive sleep.
- The positive coefficient for Quality of Sleep (3.3449) highlights that higher self-reported sleep quality is associated with greater productivity.

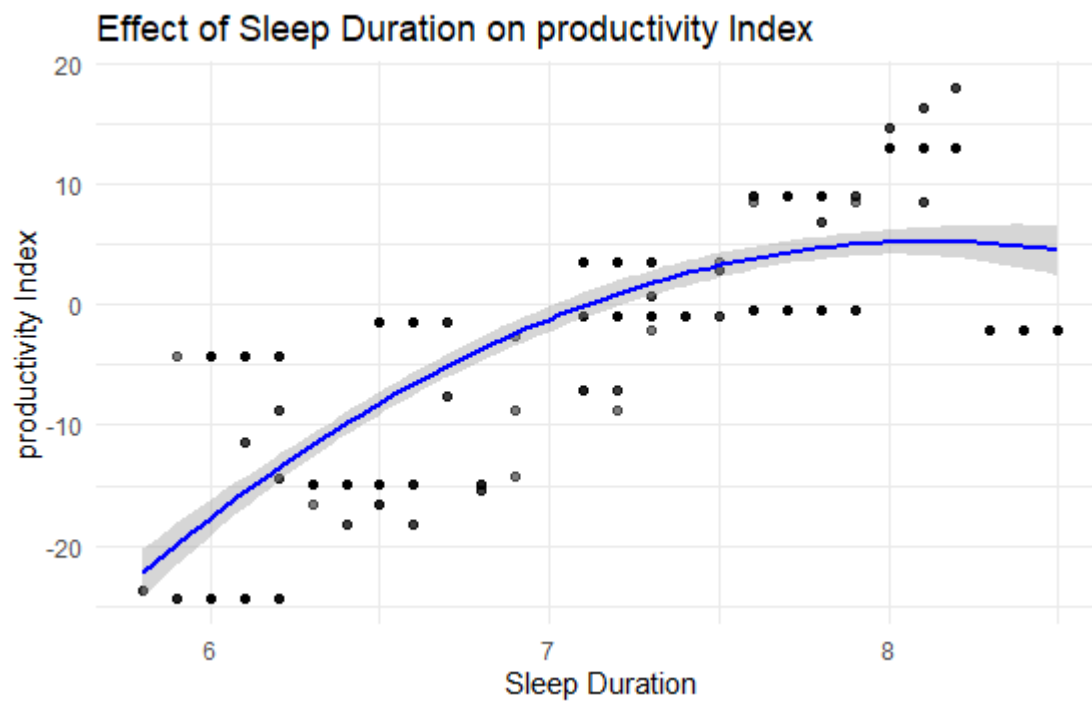
The Residual Standard Error of 5.79 on 369 degrees of freedom indicates that the model fits the data reasonably well, with residuals displaying random variation around zero. Diagnostic checks confirmed that the assumptions of linearity, homoscedasticity, and normality of residuals were adequately satisfied.

Overall, this model effectively captures the combined influence of demographic and sleep-related variables on productivity, providing a strong predictive framework for understanding how lifestyle choices can enhance or reduce individual performance.

3. Software and Implementation

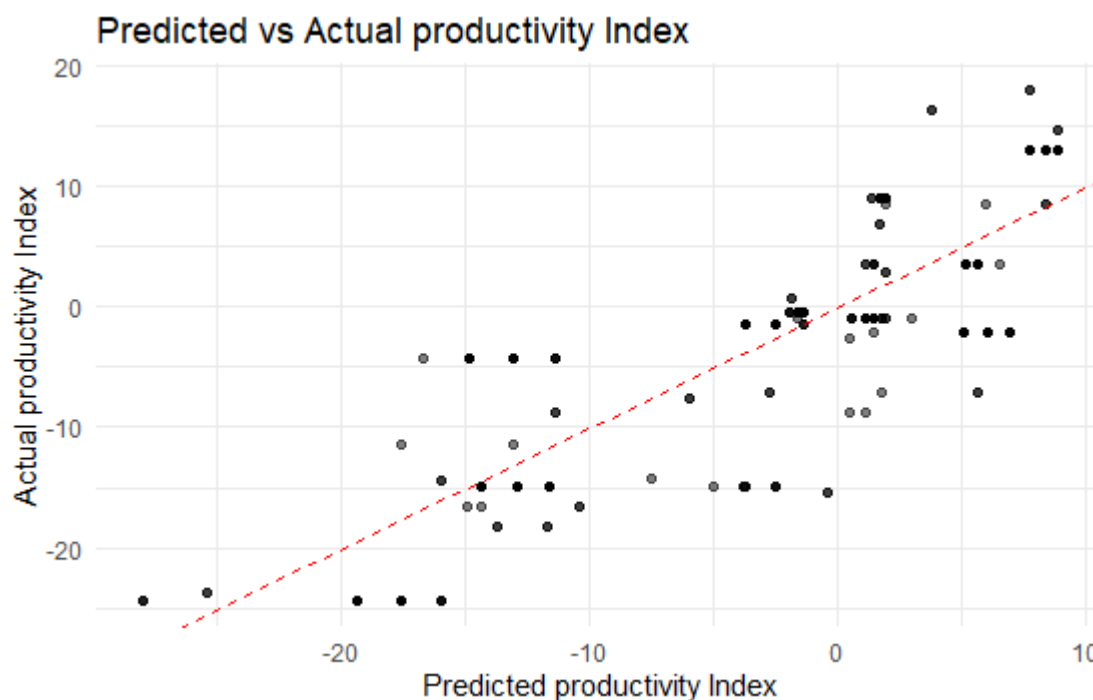
All data handling, statistical modeling, and visualization were performed using the R programming language. The base R functions `glm()` and `lm()` were used for model fitting, while the `ggplot2` package supported data visualization. The analysis workflow ensured reproducibility, transparency, and statistical validity throughout the process.

Results & Discussion

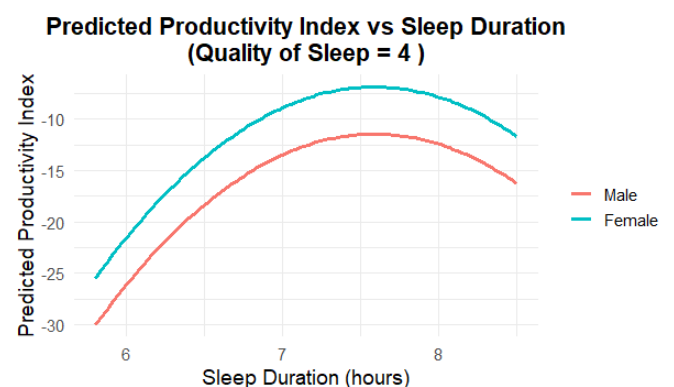
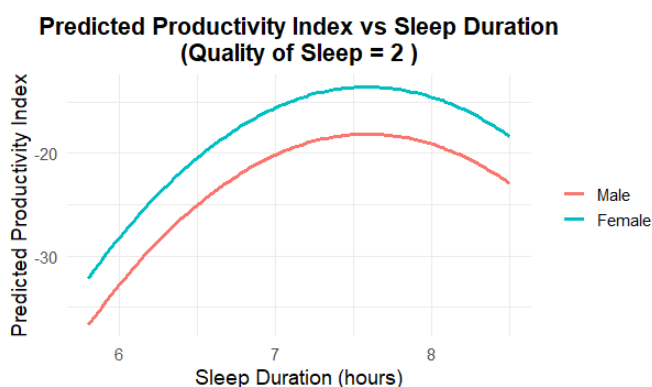


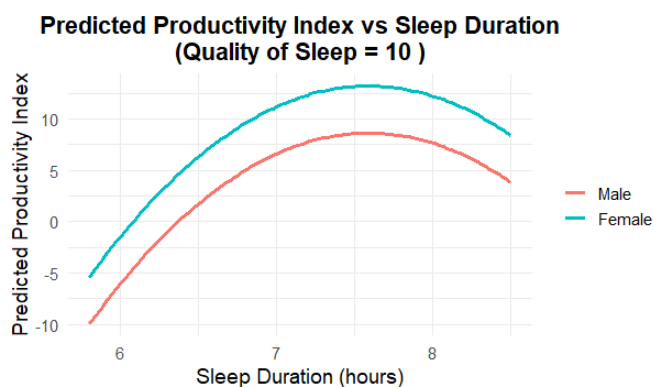
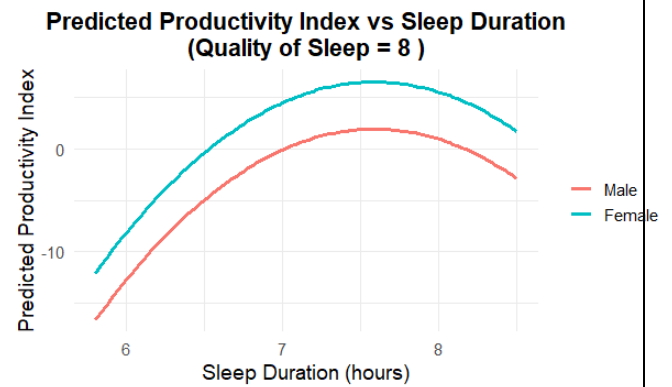
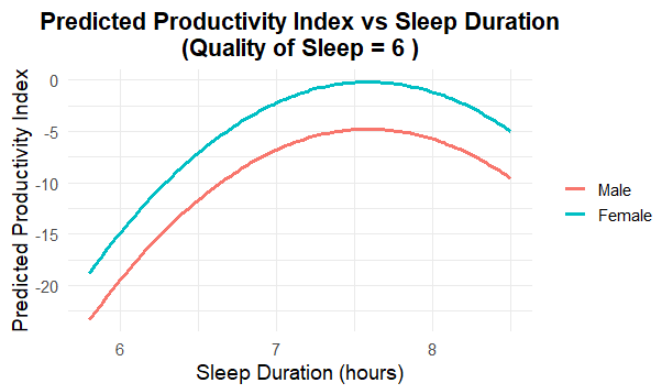
The following graph illustrates the comparison between the **actual productivity values** and the **predicted productivity values** obtained from the linear regression model. Each point represents an individual, showing whether the model's predicted classification (productive or not productive) aligns with the actual observed value.

From the graph, it can be observed that the predicted values closely follow the actual values, indicating that the model has captured the underlying relationship between physical activity, stress level, and productivity effectively. Minor deviations between actual and predicted classifications suggest that while the model performs well overall, there may be additional factors influencing productivity that are not fully represented in the current variables.



Following plots illustrates the relationship between **sleep duration** and the **predicted productivity index (PI)** for both males and females, holding the **quality of sleep constant** at a fixed level





The curves reveal that productivity initially increases with sleep duration, reaching an optimal range around **7–8 hours**, after which it begins to decline slightly. This pattern aligns with the quadratic nature of the fitted model, where the squared term for sleep duration captures this diminishing return effect.

Across both genders, females generally exhibit a **higher predicted productivity index** compared to males for the same sleep duration and sleep quality level. This difference reflects the negative coefficient for the GenderMale variable in the regression model, indicating that males tend to have a lower productivity index on average.

Overall, the graph supports the conclusion that **adequate sleep duration**—neither too little nor excessive—combined with **better sleep quality**, plays a vital role in enhancing productivity. The observed gender variation further highlights the importance of considering demographic differences in productivity-related behavioral studies.

Conclusion

This study aimed to develop a mathematical model that explains how sleep duration influences individual productivity, while also considering the effects of sleep quality and gender. A two-stage modeling approach was used. First, a logistic regression model was applied to estimate productivity probabilities based on physical activity level and stress level, from which a Productivity Index (PI) was derived using the logit transformation. Subsequently, a linear regression model was fitted to the PI, using sleep duration, its quadratic term, quality of sleep, and gender as predictors.

The results demonstrated that all selected variables were statistically significant. The positive coefficient for sleep duration and the negative quadratic term indicate that productivity increases with sleep up to an optimal level (approximately 7–8 hours) and then slightly declines beyond that. The quality of sleep had a strong positive effect, confirming that better sleep quality enhances productivity. The negative coefficient for males implies that, holding other factors constant, females tend to have a higher predicted productivity index.

The final model achieved a high R^2 value of 0.711, indicating a good fit between the predicted and actual productivity indices. Visualization of predicted versus actual values further validated the model's reliability. The gender-based plots highlighted consistent trends across sleep duration, supporting the quadratic relationship observed in the model.

In conclusion, the findings emphasize that balanced sleep duration and good sleep quality are key determinants of productivity. The study also reveals meaningful gender differences in productivity patterns. These insights could be useful in designing wellness programs and workplace policies aimed at improving employee performance through better sleep management.